1. Brainstorming and Ideation Phase

Script-Controlled ACL: Restrict Record Access Based on Field Value

Problem Statement

In a ServiceNow instance, every user with basic table access can normally see all records in a list. For a table such as **Institution Details**, which stores student, faculty, branch and contact information, this is a risk: a user from one branch should not be able to browse records belonging to another branch. Standard role-only ACLs are too coarse, because they either grant or deny access to the whole table.

Idea

Implement a **script-controlled Access Control List (ACL)** so that record visibility depends on both the user's roles and the value of a field on the record. Only users of the **EEE branch** may view records where **Branch = EEE**, while administrators retain full access.

Brainstormed Approaches

Approach	Description	Outcome
Role-only ACL	Grant a role read access to the whole table	Rejected: no record-level control
Data-condition ACL	Add a condition such as Branch is EEE to the ACL	Selected: filters records by field value
Advanced (scripted) ACL	Use an ACL script to evaluate roles with gs.hasRole()	Selected: lets admin bypass and gives flexible logic
Separate CRUD roles	One role per operation (read, create, write, delete)	Selected: bb1 to bb4 give layered permissions

Chosen Solution

- Combine a **role requirement**, a **data condition** and an **ACL script** on the u_institution_details table.
- Use four custom roles (**bb1, bb2, bb3, bb4**) so each operation can be granted and tested independently.
- Use a test user (**EEE User**) with impersonation to prove the behaviour before and after each ACL.

Expected Benefits

- Protects sensitive student and faculty data from unauthorized viewing, modification, creation and deletion.
- Demonstrates how roles and field values are evaluated together for an access decision.
- Works consistently across forms, lists and Service Portal interfaces.

Target Users

ServiceNow learners and administrators who want hands-on experience with record-level security. The lab is designed to take about **3 hours**.